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Inferring the Nominal Molecular Mass of an Analyte from Its Electron Ionization Mass Spectrum

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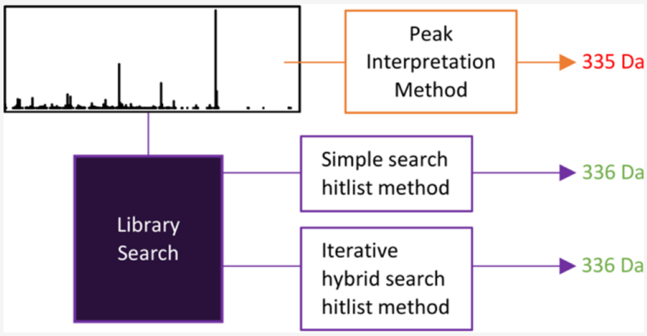
ABSTRACT: The performance of three algorithms for predicting nominal molecular mass from an analyte's electron ionization mass spectrum is presented. The Peak Interpretation Method (PIM) attempts to quantify the likelihood that a molecular ion peak is contained in the mass spectrum, whereas the Simple Search Hitlist Method (SS-HM) and iterative Hybrid Search Hitlist Method (iHS-HM) leverage results from mass spectral library searching. These predictions can be employed in combination (recommended) or independently. The methods were tested on two sets of query mass spectra searched against libraries that did not contain the reference mass spectra of the same compounds: 19,074 spectra of various organic molecules searched against the NIST17 mass spectral library and 162 spectra of small molecule drugs searched against SWGDRUG version 3.3. Individually, each molecular mass prediction method had computed precisions (the fraction of positive predictions that were correct) of 91, 89, and 74%, respectively. The methods become more valuable when predictions are taken together. When all three predictions were identical, which occurred in 33% of the test cases, the predicted molecular mass was almost always correct (>99%).



■ INTRODUCTION

Given an unidentified analyte, determining its molecular mass is a critical step toward confidently proposing its molecular structure.[1−4] Electron ionization mass spectrometry (EI-MS) is one of the most widely employed techniques to elucidate chemical structure. In EI-MS, an analyte is ionized by impact with energetic electrons and the relative intensities of its resulting ions are recorded across a mass-charge range in a mass spectrum. A molecular ion is an ion of the intact molecule, and the ions that result from fragmentation are fragment ions. If the resulting mass spectrum contains a molecular ion peak, the molecular mass of the analyte can be directly determined. However, the presence of a molecular ion peak is not guaranteed in EI-MS; thus, complimentary techniques (e.g., chemical ionization) may be required to establish the molecular mass and, eventually, the identity of the analyte.

To address this practical limitation, we present three different algorithms for predicting the nominal molecular mass of an analyte given only its EI mass spectrum and a reference library. The first algorithm attempts to quantify the likelihood that a molecular ion is contained in the mass spectrum, while the others leverage results from mass spectral library searching. These predictions can be used independently or as an ensemble (recommended) to infer the nominal molecular mass of the analyte.

■ MATERIALS AND METHODS

Mass Spectral Library Searching. Several descriptions of library search techniques employed by the National Institute of Standards and Technology (NIST), in the context of both EI-MS and electrospray ionization (ESI) MS/MS, can be readily found in the literature.[5−8] This section of the paper briefly summarizes two library search procedures necessary for describing the molecular mass prediction algorithms that follow.

All mass spectral library search procedures can be broadly described as methods to sort the library mass spectra according to similarity with a query mass spectrum. Compounds associated with the library mass spectra most similar to the query spectrum are returned as an ordered hitlist (or hit list). The results of the library search, consequently, are dependent on the content in the mass spectral library and how spectral similarity is computed.

In a Simple Similarity Search, the mass spectral similarity between a library mass spectrum and the query is measured by a simple match factor, which ranges from 0, when the

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compared mass spectra have no peaks in common, through to 999, when the compared mass spectra are identical. Values approaching, but not identical to, 999 are obtained when the compared mass spectra share several major peaks. Ideally, this means that the spectra are replicate measurements of one molecule that can then be uniquely identified. However, some isomers and structural analogs may also result in spectra with large simple match factors. We refer to pairs of unique molecules with numerically indistinguishable mass spectra as isospectral.

Another important library search procedure is the Hybrid Similarity Search. As its name implies, the mass spectral similarity between a library mass spectrum and the query is measured by a hybrid match factor. This match factor also ranges between 0 and 999; however, a value of 999 can be attained even when the mass spectra are not identical. Hybrid match factors allow for peaks from the library mass spectrum to be shifted by the molecular mass difference (known or estimated) between the compounds producing the compared spectra, DeltaMass. This means that hybrid match factors approaching and including 999 can occur between mass spectra that share matching peaks as well as non-matching peaks that can be explained through a shift by DeltaMass. Typically, these are spectra of analogs with structural modifications that affect only a single fragmentation pathway (cognates) as well as the same isospectral molecules seen with large simple match factors.

More details about simple and hybrid match factors can be found in the literature.[5−10] Several papers describing the general effectiveness of library searching are also available[5,11−13] as are several recent applications of the Hybrid Similarity Search.[8,14−19]

Methods for Predicting Nominal Molecular Mass. The first considered method is referred to as the Peak Interpretation Method (PIM). The pseudocode for this method is presented as Algorithm 1 in the Supporting Information. Principles underlying this method have been used by analytical chemists for several decades; here, they are communicated as a logical sequence of steps. The method requires only a query mass spectrum. Its prediction (output) is denoted as $\bar{m}_I$. The approach screens peaks in the mass spectrum, beginning with the highest mass peak moving toward lower mass peaks as necessary. It then identifies a potential molecular ion peak based on the following assumptions:

(i) The molecular ion peak ($M^{+\cdot}$) is contained in the spectrum.

(ii) The abundance of $M^{+\cdot}$ is greater than some minimum abundance (ϵ_M).

(iii) Peaks with mass greater than $M^{+\cdot}$ must be explainable as isotope peaks.

(iv) Nearby peaks with mass less than $M^{+\cdot}$ must not be a result of "illogical"[20,21] neutral losses.

The mass of the identified potential molecular ion peak is then the proposed molecular mass, $\bar{m}_I$, of the analyte.

We can use assumption (iv) to estimate the likelihood that the proposed molecular mass is correct. In particular, a set of estimated neutral loss masses associated with the query spectrum can be constructed with each entry of the set computed as the difference between $\bar{m}_I$ and all fragment ion masses in the spectrum. If $\bar{m}_I$ is the true molecular mass of the analyte, most estimated neutral losses should follow our expectations of logical neutral loss values.[20,21] If $\bar{m}_I$ is not the true molecular mass of the analyte, some of the estimated neutral losses may be illogical. While there are some general rules about acceptable neutral losses, these often include more nuanced ideas such as isotopic envelopes that are difficult to generalize. Accordingly, we consider a data-driven first approximation of neutral loss logicality for ease of implementation. Further refining the algorithm to incorporate known rules and more sophisticated modeling is on-going work.

Figure 1 shows the fraction of occurrence, $F(l)$, for observed loss peak, l, in the mass spectra of organic molecules with

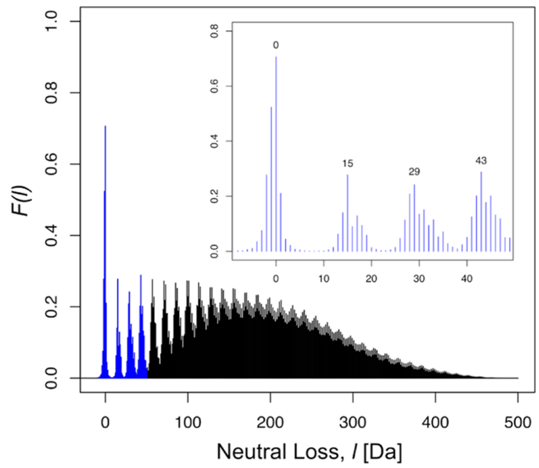

Figure 1. Fraction of the spectra from a selection of organic molecules with molecular mass between 50 and 500 Da in the NIST17 Main Library, $F(l)$, with loss peak, l, associated with peaks greater than 1% abundance. The neutral loss values highlighted in blue and inset in the figure are the most diagnostic values when computing the logicality index of eq 2. Losses such as 15, 29, 43, etc., are common and logical. Losses of 6, 10, 22, etc., are uncommon and illogical.

nominal molecular mass between 50 and 500 Da contained in the NIST17 Main Library.[22] Note that any $l = 0$ is indicative of a molecular ion peak, and $l < 0$ represent peaks of isotopes of molecular ions. All $l > 0$ are associated with neutral loss peaks or isotopes thereof. A loss peak is defined as illogical if its fraction of occurrence in the NIST17 Main Library is less than or equal to a selected threshold fraction, say τ_F.

A predicted molecular mass that results in even a single illogical loss peak is an indication of increased uncertainty. This is amplified if the intensity associated with the illogical peak is large. Additionally, a potential molecular ion peak with low intensity further confuses the situation. It is useful to examine the ratio of intensity associated with questionable loss peaks to the intensity of the molecular ion peak as a proxy for uncertainty. We define this ratio to be:

$$r(\tau_F) = \frac{\sum_{i=1}^{n_L} \max(a_i - \epsilon_F, 0)}{a_{\bar{m}_I}} \quad (1)$$

where the set $\mathbf{a} = \{a_1, a_2, ..., a_{n_L}\}$ are the intensities associated with the $n_L > 0$ questionable neutral loss masses, ϵ_F is the user-prescribed maximum intensity tolerated for an illogical peak (e.g., 1% of the base peak), and $a_{\bar{m}_I}$ is the intensity of the potential molecular ion. Employing the sigmoid function

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$$I_{\bar{m}_I}(r(\tau_F)) = 2 - \frac{2}{(1 + \exp(-\beta \cdot r(\tau_F)))} \quad (2)$$

the ratio (eq 1) can be transformed into a score taking values between 0 and 1, where β is a user-defined parameter. We refer to the index $I_{\bar{m}_I}$ as a logicality index as a value of 1 indicates that the spectrum contains no illogical neutral losses.

The second method we present is the Simple Search Hitlist Method (SS-HM). The pseudocode for this method is included as Algorithm 2 in the Supporting Information. The SS-HM seeks to improve predictions generated using the PIM of a query by considering how the PIM performs for similar mass spectra as identified through a mass spectral library search—it can be thought of as a predictor−corrector method, motivated by ideas presented by Scott[23−25] and Scott *et al.*[26] The evaluation of proposed mass corrections is built on the principles of chemical substructure identification by library searching presented by Stein.[27]

The SS-HM requires a query spectrum and reference spectral library, and its prediction is $\bar{m}_S$. The algorithm begins with a Simple Similarity Search of the query spectrum. For the top n_H entries of the hitlist, the differences between their true nominal molecular mass and predictions obtained using the PIM are computed; here, we refer to these differences as mass corrections. The collection of n unique mass corrections computed forms a set of potential mass corrections, we call c, for the PIM prediction of the analyte. Each element of the set c is then assigned an index, ξ_{c_i}, based on its associated simple match factors versus those associated with all potential mass corrections,

$$\xi_{c_i} = \frac{\sum_{j \in \mathbf{n}_{c_i}} 2^{(\mathbf{S}[j] - \mathbf{S}[1])/B}}{\sum_{k \in \mathbf{n}_H} 2^{(\mathbf{S}[k] - \mathbf{S}[1])/B}}$$

where $\mathbf{n}_{c_i}$ is defined to be the set of indices for hits associated with potential mass correction c_i, $\mathbf{n}_H = \{1,2, ..., n_H\}$, and B is a user-defined parameter representing a minimum difference in match factors deemed to be substantial. The potential mass correction with the largest index, say c^*, is the proposed correction of the PIM prediction of the analyte. The SS-HM prediction is thus $\bar{m}_S = \bar{m}_I + c^*$. A classification score for $\bar{m}_S$ is defined as $I_{\bar{m}_S} = \max(\xi_{c_i})$.

The final method presented in this paper is the iterative Hybrid Search Hitlist Method (iHS-HM). The pseudocode for this method is included as Algorithm 3 in the Supporting Information. This method requires a query spectrum, a reference spectral library, and a set of potential nominal molecular masses for the analyte and produces an output denoted by $\bar{m}_H$.

For each potential molecular mass, a hybrid search is executed with DeltaMass values approximated using the potential molecular mass of the analyte. Entries in the hitlist with hybrid match factors greater than a minimum expected match factor (mEMF) and with non-zero DeltaMass are designated potential cognates, and hits with match factors greater than a mEMF with DeltaMass of zero are designated potential isomers. If a hitlist contains at least one potential isomer or cognate, a score elevation, denoted as Ω, is computed from the returned hybrid search hitlist

$$\Omega = \frac{1}{|N_1| + |N_0|} \sum_{j \in N_1} \left(\frac{H_j - \text{mEMF}}{1000 - \text{mEMF}} \right) (H_j - S_j) + \frac{1}{|N_1| + |N_0|} \sum_{k \in N_0} \left(\frac{H_k - \text{mEMF}}{1000 - \text{mEMF}} \right) H_k$$

where N_1 are the indices for the set of potential cognates, N_0 are the indices for the set of potential isomers, H and S are the sets of hybrid and simple match factors extracted from the hitlist, and we use the notation $|\cdot|$ to specify the number of elements in the set. If a hitlist does not contain any isomers or cognates, then $\Omega = 0$.

The assumed molecular mass resulting in a hitlist with the largest score elevation is the predicted molecular mass of the analyte, $\bar{m}_H$. The normalized difference between the first and second largest score elevation, denoted as Ω_1 and Ω_2, respectively, is used to compute a classification score, $I_{\bar{m}_H}$, for the proposed molecular mass,

$$I_{\bar{m}_H} = \frac{\Omega_1 - \Omega_2}{999}$$

Parameters. Algorithms described in this paper have parameters that must be prescribed by the user. Predicted molecular masses as well as their classification scores depend strongly on these parameters. For example, selecting a high value for the minimum expected match factor of a cognate or isomer (mEMF) could prevent true cognates/isomers from being used in Algorithm 3, but selecting low a value could introduce undesirable outcomes as well. A user should consider the quality of their query mass spectra when selecting values for Algorithm 1 and the construction of their mass spectral library when selecting values for Algorithms 2 and 3. Table 1 summarizes the preliminary parameter values employed for demonstrative examples in this paper, as well as the range of feasible values for all parameters. Optimizing these parameters for particular application areas will lead to improved performance.

■ RESULTS AND DISCUSSION

Example Computations. To illustrate the operation and efficacy of the algorithms, three different test cases are presented. All computations use the parameter values outlined in Table 1. For library-based methods (i.e., SS-HM and iHS-HM), we used the Scientific Working Group for the Analysis of Seized Drugs (SWGDRUG) MS Library Version 3.3 as the reference library. Current versions of the SWGDRUG MS Library can be obtained at https://swgdrug.org.[28]

Example 1. Consider the mass spectrum of an analyte shown in Figure 2. Using the PIM (Algorithm 1), we predicted its molecular mass to be $\bar{m}_I$ = 383 Da. With that prediction, there is only one questionable neutral loss estimate with an abundance greater than the minimum acceptable abundance of fragment ions, ϵ_F, assigned a default value of 0.0% as defined in Table 1. The relative intensity for the questionable neutral loss estimate, however, is 0.5%, and the abundance of the proposed molecular ion is 35%. The questionable peak abundance ratio determined by eq 1 is approximately 0.01, and the index $I_{\bar{m}_I}$ determined by eq 2 is computed to be 0.96.

The simple search employed by the SS-HM (Algorithm 2) yielded a hitlist with unique mass corrections and simple match factors summarized in Table 2. A large fraction (20/25) of the

Table 1. Comprehensive List of Parameters Used in Algorithms 1 through 3[c]

symbol	description	value	range
Algorithm 1: Peak Interpretation Method (PIM)			
ϵ_M	min. acceptable abundance of a potential molecular ion (percentage of base peak)	0.3	0−100
ϵ_F	max. tolerated abundance of an illogical neutral (percentage of base peak)	0.0	0−100
τ_F	min. fraction of occurrence for a peak to not be considered "questionable"	0.06	0−1
β	sigmoid function (eq 2) rate parameter	5	>0
Algorithm 2: Simple Search Hitlist Method (SS-HM)			
N	number of hitlist entries considered when searching for unique mass corrections	25[a]	>1[a]
B	difference in match factors considered substantial	75[a,b]	1−999[a,b]
Algorithm 3: iterative Hybrid Search Hitlist Method (iHS-HM)			
mEMF	minimum expected match factor of a cognate or isomer	700[a,b]	1−999[a,b]

[a]Must be an integer value. [b]Values are presented, assuming that similarity match factors are computed between 0 and 999 as described in this paper. [c]The listed default values were identified through experience (e.g., parameters for Algorithms 1 and 2) or preliminary tuning studies with subsets of spectra (e.g., parameters in Algorithm 3).

hitlist compounds had mass correction values of 0 Da, and thus, the SS-HM predicted $\overline{m}_S$ = 383 Da with $I_{\overline{m}_S}$ = 0.88.

To employ the iHS-HM, a range of potential molecular masses must be specified. We considered all integer masses between 355 and 400 Da for demonstrative purposes. Efficient strategies for selecting a set of potential mass values will vary with application. The score elevation for each potential molecular mass is presented in Figure 3. The molecular mass predicted using the iHS-HM method is $\overline{m}_H$ = 383 Da with $I_{\overline{m}_H}$ = 0.06.

All three methods predicted the same molecular mass but with different indices. Decision-making threshold values for each index are discussed in greater detail in the following. We note that the threshold values appropriate for this example (spectrum of a small molecule drug) are $\tau_{\overline{m}_I}$ = 0.01, $\tau_{\overline{m}_S}$ = 0.65, and $\tau_{\overline{m}_H}$ = 0.01 for $I_{\overline{m}_I}$, $I_{\overline{m}_S}$, and $I_{\overline{m}_H}$, respectively (see Table 4).

Table 2. Summary of Simple Search Results Used to Generate SS-HM Prediction for Example 1[a]

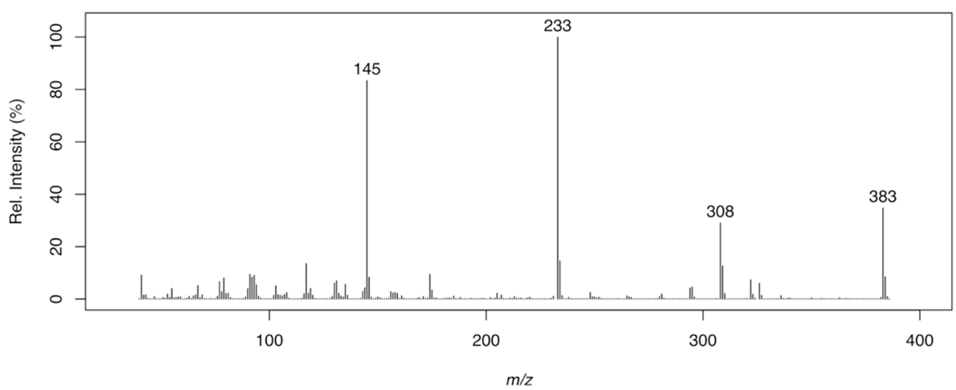
c_i	number of hits	Simple Match Factors of hits with listed correction				ξ_{c_i}
0	21	688	609	···	424	0.88
57	1	532				0.05
2	1	465				0.03
31	1	441				0.02
18	1	429				0.02

[a]Each unique mass correction obtained from the hitlist is shown in the column labeled c_i with the number of occurrences the next column. Not all 21 simple match factors associated with c_i = 0 were listed to save space. The index associated with each mass correction value is listed in the right-most column labeled ξ_{c_i}.

All three indices for Example 1 exceed their corresponding threshold values, suggesting that the predictions are the correct molecular mass of the analyte. The molecule producing Example 1 is 5-fluoro-7-APAICA ($C_{23}H_{30}FN_3O$), and the true molecular mass is 383 Da.

Example 2. The molecule producing the spectrum in Figure 4 is *N*-methylethylone ($C_{13}H_{17}NO_3$) with a nominal molecular mass of 235 Da. As there is no molecular ion in this spectrum, the PIM predicted a molecular mass of $\overline{m}_I$ = 160 Da. However, this prediction yields several questionable neutral-loss estimates such that $I_{\overline{m}_I}$ = 0. The mass spectra in the reference mass spectral library most similar to the query had mass corrections of 29 Da, and so, the SS-HM predicted a mass of $\overline{m}_S$ = 189 Da with $I_{\overline{m}_S}$ = 0.25. Note that if the query spectrum (or a reference spectrum of the query compound) was included in the library, then the SS-HM prediction would have likely been different. The iHS-HM, with a range of 200 to 270 Da, predicted a molecular mass of $\overline{m}_H$ = 235 Da with $I_{\overline{m}_H}$ = 0.22. In this example, only the index associated with the iHS-HM prediction was greater than its recommended threshold value; thus, an analyst should correctly infer the molecular mass.

Example 3. The molecule producing the spectrum shown in Figure 5 is *para*-fluoro cyclopentylfentanyl ($C_{25}H_{31}FN_2O$) with a true molecular mass of 394 Da. Here, the PIM incorrectly predicted a molecular mass of $\overline{m}_I$ = 392 Da with $I_{\overline{m}_I}$ = 1.0. The SS-HM predicts a molecular mass of $\overline{m}_S$ = 394 Da with $I_{\overline{m}_S}$ = 0.43. The iHS-HM, with a range of 359 to 429 Da, predicts a molecular mass of $\overline{m}$ = 394 with $I_{\overline{m}_I}$ = 0.14. In this



10 Largest Peaks

m/z	Rel. Intensity [%]
233	100
145	83
383	35
308	29
234	15
117	14
309	13
91	9
174	9
41	9

Figure 2. Complete mass spectrum and summary of 10 most abundant fragment ions for analyte considered in Example 1.

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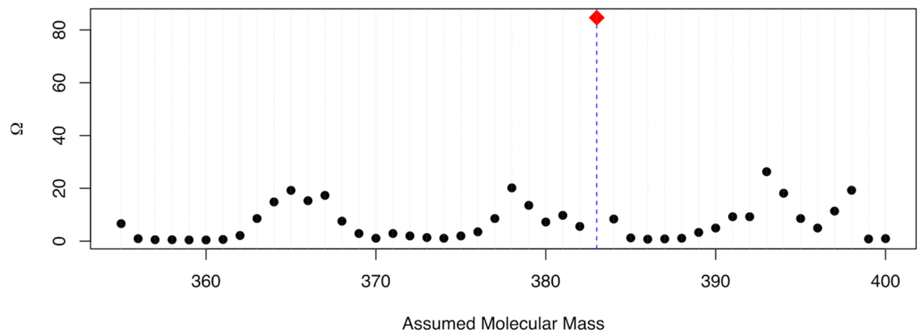
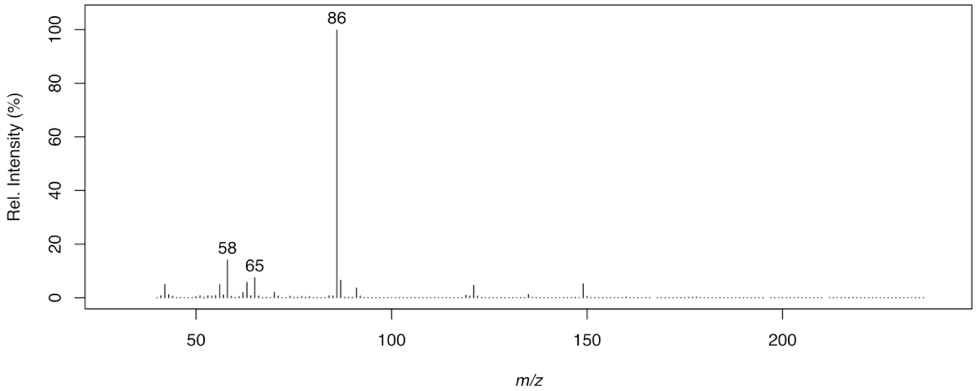


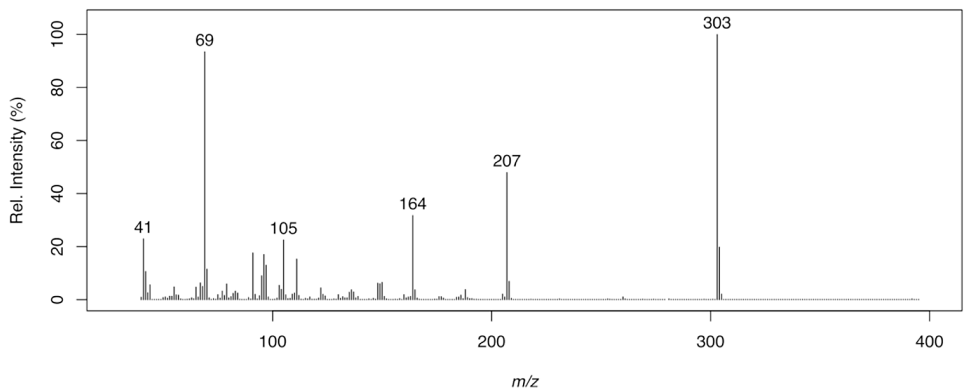
Figure 3. Results of using the iterative Hybrid Search Hitlist Method (iHS-HM) to predict the molecular mass of the analyte shown as Example 1. The method correctly identifies 383 Da as the molecular mass of the query compound.



10 Largest Peaks

m/z	Rel. Intensity [%]
86	100
58	14
65	8
87	7
63	6
149	5
42	5
56	5
121	5
91	4

Figure 4. Complete mass spectrum and summary of 10 most abundant fragment ions for analyte considered in Example 2.



10 Largest Peaks

m/z	Rel. Intensity [%]
303	100
69	93
207	48
164	32
41	23
105	23
304	20
91	18
96	17
111	15

Figure 5. Complete mass spectrum and summary of 10 most abundant fragment ions for analyte considered in Example 3.

example, an analyst may have difficulty determining the molecular mass of the analyte as both $I_{\bar{m}_I}$ and $I_{\bar{m}_H}$ are above their threshold values, yet the two library-based predictions are corroborative. It should be noted that if an analyst suspected that the analyte was a fentanyl analog, they may have justified the difference between $\bar{m}_I$ and the library-based predictions, as 2 Da is a commonly observed loss among this class of compounds. As noted in the description of the PIM algorithm, the logicality index $I_{\bar{m}_I}$ summarizes the number of questionable peaks based on estimated neutral loss rather than the likelihood of the predicted molecular mass being correct, which thus should be treated differently by an analyst.

Classification Thresholds and Decision-Making. A summary of the prediction results for the previous examples is provided in Table 3. The indices for each method vary significantly in their values and utility toward classification and decision-making. To characterize the performance of computed indices as binary classifiers, two independent performance evaluations were conducted. The first study considered a set of 19,074 spectra of organic molecules (including elements C, H, O, N, S, P, F, Cl, Br, and I) with molecular masses between 50 and 500 Da, inclusive. The NIST17 Main Library was used for library search-based predictions in this study. The second study considered 162 spectra of small molecule drugs.

Table 3. Summary of Molecular Mass Predictions for Examples Presented in the Paper

ex.	true mass	$\bar{m}_I$ ($I_{\bar{m}_I}$)	$\bar{m}_S$ ($I_{\bar{m}_S}$)	$\bar{m}_H$ ($I_{\bar{m}_H}$)
1	383 Da	383 (0.96)	383 (0.88)	383 (0.06)
2	235 Da	160 (0)	189 (0.25)	235 (0.22)
3	394 Da	392 (1)	394 (0.43)	394 (0.14)

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Library search-based predictions in this study employed the SWGDRUG MS Library v3.3. We did not consider the spectra of derivatized compounds in either test set.

Each of the computed indices ($I_{\bar{m}_I}$, $I_{\bar{m}_S}$, and $I_{\bar{m}_H}$) takes values between 0 and 1. Let τ_γ, between 0 and 1, be the decision thresholds for a given index I_γ. That is, when $I_\gamma > \tau_\gamma$, we assume that the predicted molecular mass γ is the true molecular mass of the analyte, and when $I_\gamma \le \tau_\gamma$, we assume that the predicted γ is incorrect. For a given τ_γ and set of test data, we can generate a contingency table, commonly known as a confusion or error matrix, that summarizes how well the classifier threshold performs under tested circumstances,

conditions	$\gamma = MW$	$\gamma \ne MW$
$I_\gamma > \tau_\gamma$	TP	FP
$I_\gamma \le \tau_\gamma$	FN	N

where MW is the true molecular mass, and the symbols TP, FP, FN, and TN indicate the number of true positives, false positives, false negatives, and true negatives, respectively, observed in the test. From the error matrix, several common performance metrics can be calculated.[29] The metrics considered in this paper are summarized in Table S1. Of most importance is the measure of precision, which estimates the probability that a prediction with an index greater than the threshold is correct.

A summary of classification performance metrics on the outlined tests sets is provided in Table 4, with the threshold value, denoted as $\tau\_$ for generality, chosen to maximize the computed accuracy of each method. In general, these results demonstrate that all three methods can be used independently with the prescribed threshold values to predict the molecular mass of a molecule from the EI mass spectra even without parameter optimization as discussed earlier in the paper. Analysis of standard receiver operating characteristic curves for a range of threshold values is provided in the Supporting Information. Figure 6 shows the standard receiver operating characteristic (ROC) curves for all three algorithms using test set 1 (19,078 predictions), with threshold values assessed between 0.001 and 0.999 by increments of 0.001. The ROC curve for an ideal classifier would have a sharp point near the top left corner, indicating that the method had high specificity and recall—the plots in Figure 6 make it clear that all three classifiers show room for improvement.

Table 4. Global Performance Results[a]

method	$\tau\_$	accuracy	precision	recall
Test set 1 (19,074 spectra of organic molecules)				
PIM	0.01	0.842	0.914	0.906
SS-HM	0.43	0.884	0.892	0.987
iHS-HM	0.02	0.730	0.741	0.756
Test set 2 (162 spectra of small molecule drugs)				
PIM	0.01	0.741	0.738	0.849
SS-HM	0.65	0.784	0.845	0.804
iHS-HM	0.01	0.660	0.623	0.931

[a]The symbol $\tau\_$ denotes a classification index threshold value chosen to maximize the returned accuracy. Performance measures computed following procedures outlined by Fawcett.[29] Definition of performance measures included as the Supporting Information (Table S1).

Of the three methods, the overall worst performing method in the large-scale studies was the iHS-HM. However, as shown in the examples, there are scenarios where the iHS-HM correctly predicts molecular weight, and the other two methods fail. Accordingly, considering all three predictions simultaneously is a useful and recommended strategy.

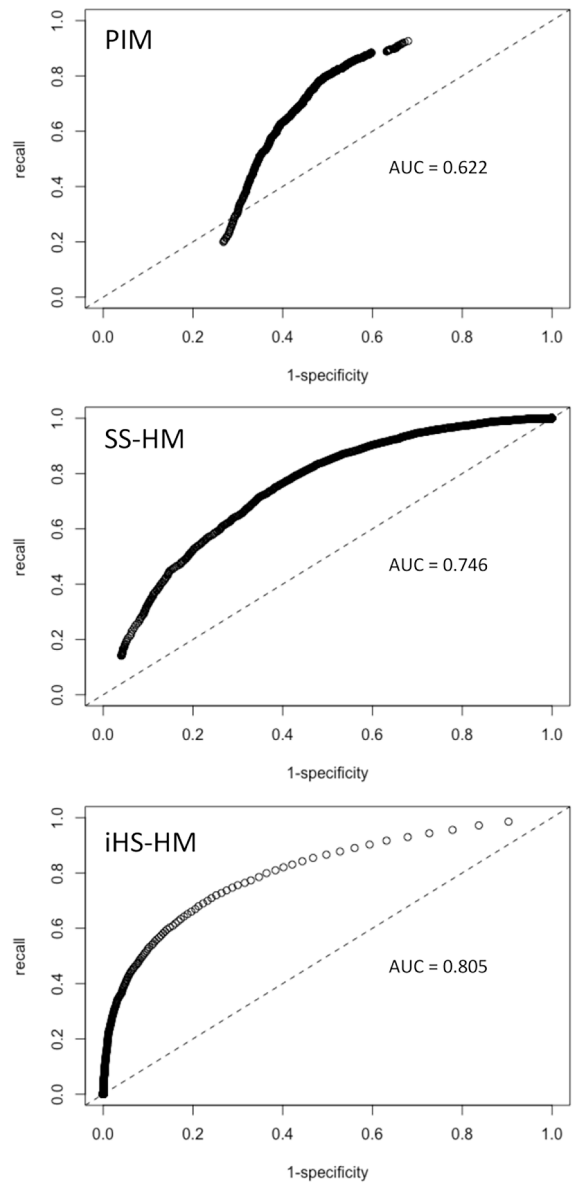

Figure 6. Receiver operating characteristic (ROC) curves illustrating the performance of binary classification using indices generated with the three methods described in this paper. Indices generated using the iterative Hybrid Search Hitlist Method (iHS-HM) demonstrated behavior most similar to the expected ideal; however, predicted molecular masses were most accurate using optimally selected thresholds with the Peak Interpretation Method (PIM) and Simple Search Hitlist Method (SS-HM) as summarized in Table 4.

We refer to situations when two or more methods produce the same prediction as ensemble predictions. Of the 19,074 test spectra in set 1, there were 18,734 for which ensembles of two or more of the predictions agreed. Of the 162 examples in test set 2, there were 118 ensemble predictions. Table 5 summarizes the results of the 32 ensemble scenarios that may arise for test set 1; Table S2 in the Supporting Information summarizes the results for test set 2. There are too few occurrences of many scenarios listed in either table to make general comments about ensemble prediction performance. However, the results suggest that having equivalent predictions

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Table 5. Summary of Ensemble Prediction Performance as Measured by Precision for the Test Set of 19,074 Spectra of Organic Molecules[a]

corroborative prediction scenario	classifier values			occurrence	precision
	$I_{\bar{m}_I} > \tau_{\bar{m}_I}$	$I_{\bar{m}S} > \tau_{\bar{m}_S}$	$I_{\bar{m}_H} > \tau_{\bar{m}_H}$		
$\bar{\mathbf{m}}_{IS} = \bar{m}_I = \bar{m}_S$, $\bar{\mathbf{m}}_{IS} \neq \bar{m}_H$	−	−	−	33	0.333
	+	−	−	106	0.642
	−	+	−	877	0.597
	−	−	+	15	0.267
	+	+	−	5200	0.870
	+	−	+	53	0.528
	−	+	+	551	0.466
	+	+	+	2334	0.813
$\bar{\mathbf{m}}_{IH} = \bar{m}_I = \bar{m}_H$, $\bar{\mathbf{m}}_{IH} \neq \bar{m}_S$	−	−	−	1	1.000
	+	−	−	12	1.000
	−	+	−	1	1.000
	−	−	+	4	1.000
	+	+	−	11	0.909
	+	−	+	15	1.000
	−	+	+	11	1.000
	+	+	+	37	1.000
$\bar{\mathbf{m}}_{SH} = \bar{m}_S = \bar{m}_H$, $\bar{\mathbf{m}}_{SH} \neq \bar{m}_I$	−	−	−	1	1.000
	+	−	−	4	0.750
	−	+	−	2	1.000
	−	−	+	1	1.000
	+	+	−	16	0.938
	+	−	+	10	1.000
	−	+	+	11	1.000
	+	+	+	30	0.967
$\bar{\mathbf{m}}_{ISH} = \bar{m}_I = \bar{m}_S = \bar{m}_H$	−	−	−		
	+	−	−	27	0.926
	−	+	−	175	0.966
	−	−	+	8	1.000
	+	+	−	2060	0.983
	+	−	+	38	0.974
	−	+	+	591	0.990
	+	+	+	6499	0.997

[a]When $I_\gamma \leq \tau_\gamma$ for the various predictions, the scenario is marked (−), as opposed to (+) when $I_\gamma > \tau_\gamma$. True positives are situations when the ensemble prediction is correct, and false positives are when the ensemble prediction is incorrect, yielding the precision included in the last column. Note: $\tau_{\bar{m}_I} = 0.01$, $\tau_{\bar{m}_S} = 0.65$, and $\tau_{\bar{m}_H} = 0.01$.

across multiple methods can be a powerful tool for an analyst. This is particularly true when all three predictions are the same, regardless of the classifier index values. When classifier values were greater than their corresponding thresholds and all three predictions agreed, the predicted molecular mass was correct, save a few instances where the labeling of the query spectrum itself can be brought into question (e.g., the analyte molecule may have degraded at the inlet).

The test sets of spectra considered in this study were taken from newer editions of the NIST and SWGDRUG MS libraries to ensure that the spectra were not contained in the reference libraries used in the SS-HM and iHS-HM. This means that the query spectra were analyzed and deemed acceptable for inclusion in high-quality libraries. It is unclear how well the molecular mass prediction methods outlined in this paper will translate to data that has been obtained in less-than-ideal circumstances. It is also unclear how well the methods will perform with the mass spectra of derivatized compounds or when using lower quality libraries. It is likely that the values of algorithm parameters (e.g., minimum expected match factor of isomers/cognates and number of hit list entries considered) will need to vary significantly from those listed in Table 1 to accommodate these less-than-ideal circumstances. Evaluating algorithm robustness to non-ideal inputs (query spectra, libraries, etc.) would be valuable future work. That said, we always recommend using the highest quality query mass spectra and mass spectral libraries, when possible, whether performing traditional library searching or computing molecular mass predictions.

■ CONCLUSIONS

This paper described three methods for predicting the molecular mass of an analyte from its EI-MS spectrum, outlining their individual and combined utility toward inferring the molecular mass of an analyte. The precision of ensemble predictions was computed for 32 scenarios that consider various classifier values. While some scenarios were inadequately represented in the results to make general performance comments, the predicted molecular mass was almost always correct when all three predictions agreed. Continuing to expand and evaluate reference libraries will improve the performance of library-based predictions. Optimizing search parameters for specific applications and improving classification of predictions for decision-making are interesting and ongoing research.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.3c01815.

Pseudocode for algorithms and supporting results (PDF)

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Notes

The authors declare no competing financial interest.
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